# HOJUN JEONG | 정호준

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## RESEARCH INTERESTS

Post-Training Knowledge Dynamics in Large Language Models; Knowledge Editing; Catastrophic Forgetting; Sequential and Continual Learning; Representation Dynamics and Mechanistic Interpretability

## EDUCATION

**Gachon University — M.S. in AI Software, Artificial Intelligence** Mar. 2025–Present
*Advisor: Prof. Sangwoo Kang | GPA: 4.3 / 4.5*

**Gachon University — B.S. in AI Software, Artificial Intelligence** Mar. 2023–Feb. 2025
*GPA: 3.6 / 4.5*

**Dongguk University — Coursework in Computer Engineering** Mar. 2018–Feb. 2023
*Transferred to Gachon University*

## MANUSCRIPTS UNDER REVIEW

*Hojun Jeong, *Gyunyeon Kim, and Sangwoo Kang. **"KLOD: Locality-Preserving Knowledge Editing via Non-Target Distribution Preservation."** Under review at ACL Rolling Review, May 2026 cycle.

*Hojun Jeong, *Gyunyeon Kim, and Sangwoo Kang. **"Target-Token Prior as a Predictor of Forgetting in Fine-Tuning-Based Knowledge Editing."** Under review at ACL Rolling Review, May 2026 cycle.

** Equal contribution.*

## CONFERENCE PUBLICATIONS

Hojun Jeong and Sangwoo Kang. **"A Verify-Refine Hybrid for Dialogue Summarization with Topic Segmentation and Keypoint Coverage."** In Proceedings of the 37th Annual Conference on Human and Language Technology (HCLT 2025), pp. 359–364, October 2025. *[Paper in Korean]*

Hojun Jeong and Sangwoo Kang. **"Structured Zero-Shot Prompts through Rule-Based Intent Analysis for Dialogue Summarization."** In Proceedings of the Korea Computer Congress (KCC 2025), July 2025. *[Paper in Korean]*

## RESEARCH EXPERIENCE

**Graduate Research Student, NLP / AI Lab, Gachon University** Mar. 2025–Present
*Supervisor: Prof. Sangwoo Kang*

**KLOD: Locality-Preserving Knowledge Editing via Non-Target Distribution Preservation**

- Identified an objective-level mismatch in standard cross-entropy editing: target probability can increase while non-target token distributions drift, causing accumulated locality degradation under sequential updates.
- Proposed KLOD, combining bounded target logit-odds optimization with KL preservation of the target-excluded non-target distribution and the full next-token distribution at prefix positions.
- Led method development, implementation, baseline selection, experimental design, ablation studies, and analysis across two backbones, two benchmarks, 3,000-edit sequential protocols, and scaling experiments up to 150,000 edits.
- On LLaMA3-8B-Instruct with CounterFact, increased Locality from 1.87 (LocFT-BF) to 44.75 while maintaining 99.70 Reliability; on ZsRE, achieved 86.47 Locality with 99.89 Reliability.
- Ablation and distribution-KL analyses identified target-excluded non-target preservation as the main source of locality gains and confirmed substantially reduced output drift; drafted the initial manuscript structure and core technical content as first-listed co-first author.

### Target-Token Prior as a Predictor of Forgetting in Fine-Tuning-Based Knowledge Editing

- Extended an initial diagnostic observation into a controlled study of target-token prior as a predictor of forgetting in fine-tuning-based knowledge editing.
- Built a synthetic biography study with 50,000 pre-training and 50,000 fine-tuning individuals, stratified into five prior-based bins, and showed that target-token NLL separated collapse trajectories more consistently than generic output confidence.
- Proposed PLFT-KE, a two-stage prior-aware warm start that optimizes a latent probability lift on a frozen model, materializes it as a rank-1 parameter initialization, and consolidates the edit with standard fine-tuning.
- Led post-diagnosis method development, implementation, baseline selection, experimental design, and analysis on CounterFact and ZsRE with Qwen2.5-7B and LLaMA-3.2-1B/3B.
- On LLaMA-3.2-3B with CounterFact, improved FT-M from 58.43 to 72.41 overall Score, including Generalization from 41.75 to 57.83 and Specificity from 33.55 to 59.40; gains reached 31.4 specificity points in the lowest-prior bin.
- Stress-tested the method over 2,000 sequential edits, identified limits under accumulated updates that motivated KLOD, and co-wrote the manuscript as first-listed co-first author.

### Ongoing Research: Early-Stage Forgetting and Layer-Wise Transformation

- Investigating why severe forgetting is concentrated in the earliest stages of fine-tuning, even under distribution-preserving objectives.
- Analyzing layer-wise transformations from rotational and scaling perspectives to characterize representation changes during post-training.
- Examining links among representation subspaces, predictive distributions, and computational pathways used to express knowledge.

**Research Assistant, NLP / AI Lab, Gachon University** Sep. 2023–Feb. 2025

*Supervisor: Prof. Sangwoo Kang*

- Conducted research on dialogue summarization and knowledge editing for large language models.
- Implemented fine-tuning, prompting, and evaluation pipelines using PyTorch and Hugging Face Transformers.
- Participated in manuscript preparation, research presentations, and regular laboratory seminars.

## AWARDS AND HONORS

**Outstanding Presentation Paper Award, HCLT 2025, October 2025.** Awarded for “A Verify-Refine Hybrid for Dialogue Summarization with Topic Segmentation and Keypoint Coverage.”

## PRESENTATIONS

**“A Verify-Refine Hybrid for Dialogue Summarization with Topic Segmentation and Keypoint Coverage.”** Oral Presentation, HCLT 2025, October 2025.

**“Structured Zero-Shot Prompts through Rule-Based Intent Analysis for Dialogue Summarization.”** Poster Presentation, KCC 2025, July 2025.

## TECHNICAL SKILLS

**Programming:** Python; working familiarity with Java and C
**Frameworks and Tools:** PyTorch; Hugging Face Transformers; EasyEdit; Git; Linux; LaTeX
**Research Methods:** Supervised and Localized Fine-Tuning; Knowledge Editing; Sequential Knowledge Editing; KL-Regularized Training; Token-Level Probability and Output-Distribution Analysis; Ablation Study Design; Language-Model Evaluation
**Languages:** Korean (native); English (academic reading and writing)